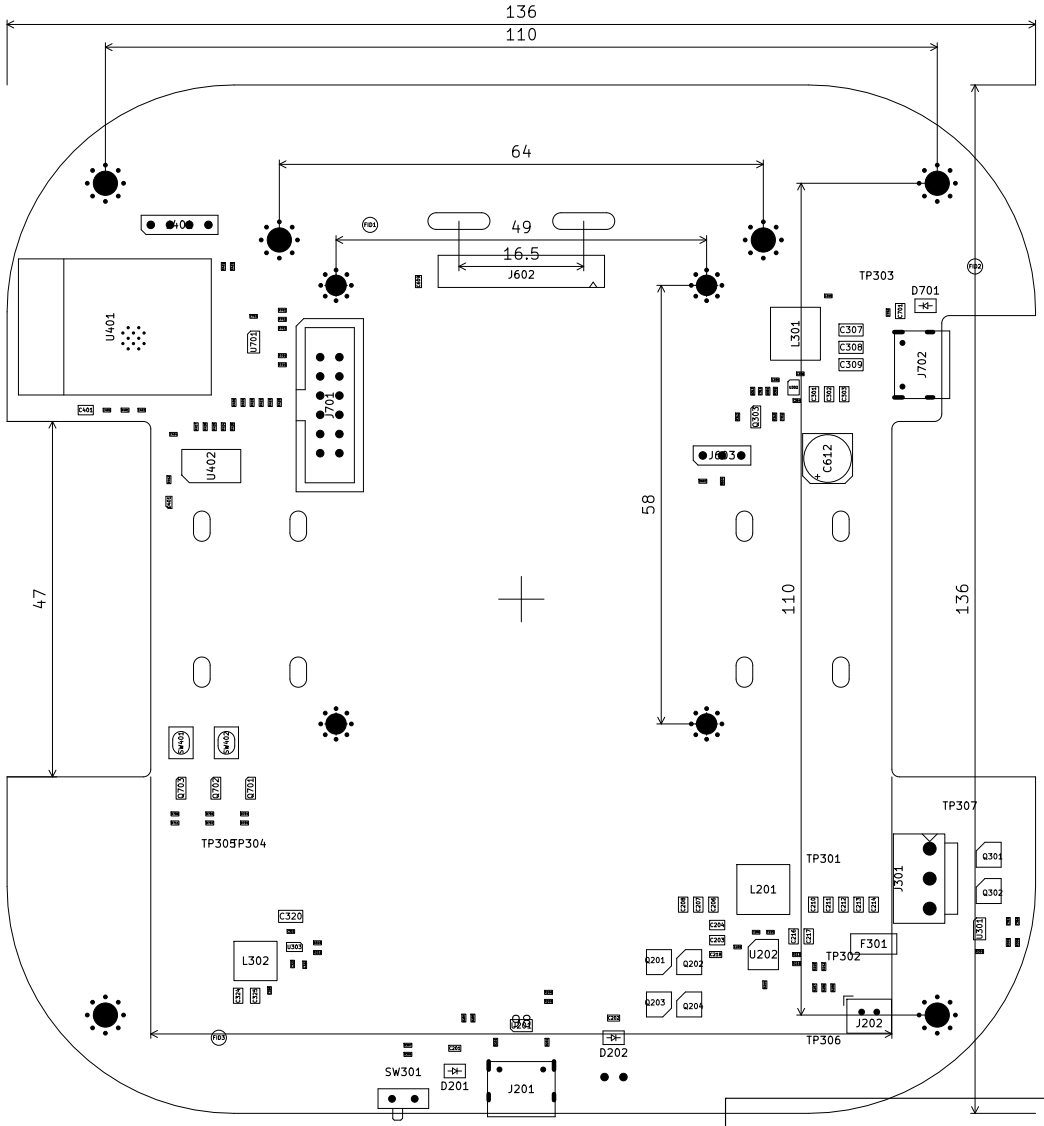


MAIN BOARD REV B — MECHANICAL DRAWING 1:1 (mm)

Origin: board centre = wheel axle. +x to the robot's right, +y towards the rear (front = −y).
136 × 136 rounded square (R30); wheel cut-outs |x| ≥ 49 for |y| ≤ 23.5; J702 plug notch on the right edge (x ≥ 55.6, y −37.5…−23.5).
M3 (ring to GND): H1-H4 shell bosses (±55, ±55); H9/H10 head angle brackets (±32, −47.5). M2.5 (ring to GND): H5-H8 Raspberry Pi 5 standoffs (45 mm: 25 + 20).
N20 bracket slots (2.2 × 4, M2): x = ±29.5 and ±42.25; y = ±9.65. Caster slots (2.2 × 8.2): (±8.25, −50).
2S2P pack on top in its cage: x ±19.4, y −37.8…30. Head board: y −53.6…−52, 76 wide (vertical).
Bottom side: N20 motors, TB6612, IMU at the axle, VL53L4CD cliff sensors (±34, −62), J203 dock lead.
Check against the real parts before fabrication: N20 brackets, caster, pack cage, Pi standoffs and angle brackets.

ronmart · Caril Ronnie Martínez Vera

Sheet:
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Title: NOVA — Main board (chassis)

Size: A4	Date: 2026-10-28	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 1/1